**Unit:** 6 Unit 8: Quadratics
**Topic:** Systems of Linear and Quadratic Equations

**Name:** ____________

**Date:** ____________

# Advanced (Habanero)

**Q1.** Solve the system: $x + y = 2$, $x^2 + y^2 = 4$
- (A) $x = 0, y = 2$
- (B) $x = 1, y = 1$
- (C) $x = 2, y = 0$
- (D) $x = -1, y = 1$
- (E) No solution

**Q2.** Find the intersection point of $y = x^2 - 2x + 1$ and $y = x - 1$
- (A) $(0, -1)$
- (B) $(1, 0)$
- (C) $(1, -1)$
- (D) $(2, 1)$
- (E) $(0, 0)$

**Q3.** Solve: $x^2 + 3x + 2 = 0$, $y = x + 1$
- (A) $x = -1, y = 0$
- (B) $x = -2, y = -1$
- (C) $x = 0, y = 1$
- (D) $x = 1, y = 2$
- (E) $x = -1, y = -1$

**Q4.** Find the solution set of $y = x^2 - 4$, $y = 2x - 4$
- (A) $x = 0, x = 2$
- (B) $x = -2, x = 2$
- (C) $x = 1, x = 3$
- (D) $x = -1, x = 1$
- (E) $x = 0, x = 4$

**Q5.** Solve the system: $x^2 + y^2 = 4$, $x - y = 0$
- (A) $x = 0, y = 0$
- (B) $x = 1, y = 1$
- (C) $x = 2, y = 2$
- (D) $x = -1, y = -1$
- (E) $x = sqrt2, y = sqrt2$

**Q6.** Find the intersection points of $y = x^2$ and $y = x + 1$
- (A) $(0, 0)$, $(1, 1)$
- (B) $(-1, 1)$, $(1, 2)$
- (C) $(-1, 0)$, $(1, 0)$
- (D) $(0, 1)$, $(1, 1)$
- (E) $(-1, -1)$, $(1, 1)$

**Q7.** Solve the system: $x^2 - y^2 = 0$, $x + y = 2$
- (A) $x = 1, y = 1$
- (B) $x = 2, y = 0$
- (C) $x = 0, y = 2$
- (D) $x = -1, y = -1$
- (E) $x = -2, y = 0$

**Q8.** Find the solution set of $y = x^2 - 1$, $y = x - 1$
- (A) $x = 0, x = 2$
- (B) $x = -1, x = 1$
- (C) $x = 1, x = 3$
- (D) $x = 0, x = 1$
- (E) $x = -2, x = 0$

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**Open Ended Questions (FRQ)**

**Q9.** Solve the system: $x^2 + y^2 = 13$, $x - y = 2$. Show all work and explain your solution.

**Q10.** Find the intersection points of $y = x^2 - 3x + 2$ and $y = x - 2$. Explain your method and show all work.

**Chef's Tips**



- Tip 1: Remind students to check their solutions by plugging them back into the original equations.
- Tip 2: Encourage students to use graphing tools to visualize the systems and check their solutions.
- Tip 3: Emphasize the importance of showing all work and explaining the method used to solve the system.